RAMAIAH Institute of Technology

Dept. of Computer Science and Engineering (AI & ML) & Computer Science and Engineering (Cyber Security)

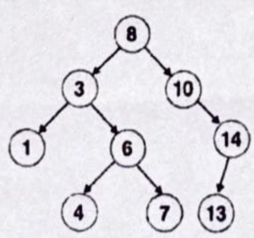
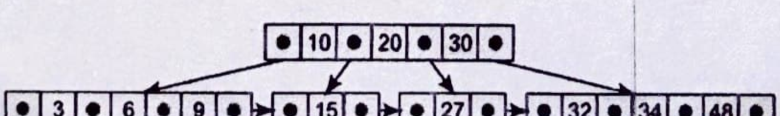
Programme: B E – Computer Science and Engineering (AI&ML) & Computer Science and Engineering (Cyber Security)

Internal Assessment – II

TERM: 3rd October 2024 to 25th Jan 2025	**COURSE NAME:** Data Structures
DATE: 21/01/2025 11:00AM to 12:00PM	**COURSE CODE :** CI33/CY33
MAX MARKS: 30	**PORTIONS :** L18 to L42

Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.** Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Write pseudocode to perform the following operations on singly linked list. i. insertion in front ii. deletion from rear	L2	CO3	5
b	Construct a binary tree by using the following traversals. In–order Traversal: D B H E I A F J C G Post order Traversal: D H I E B J F G C A	L3	CO4	5
c	What is collision in Hashing? Explain Chaining collision resolution technique with suitable example.	L2	CO5	5
2.a	Write pseudocode to perform the following operations on singly circular linked list. i. insertion in the end. ii. deletion from a specific location	L2	CO3	5
b	Explain the importance of balance factor in AVL trees. Construct AVL tree for the elements given below, 25, 45,10,18,16,50,65	L3	CO4	5
c	Construct a B tree of order 5 by inserting the following elements: 3, 14, 7, 1, 8, 5, 11, 17, 13, 6, 23, 12, 20, 26.	L3	CO5	5
3.a	Write a C function to reverse a singly linked list.	L3	CO3	5
b	Perform the preorder and postorder traversals for the given binary search tree in Fig.3.b  Fig.3.b Binary Tree	L3	CO4	5
c	Consider the B+ tree of order 4 given in Fig 4.c and insert 33 in it.  Fig 4.C	L3	CO5	5

* **L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create